

HDOC – 21th day

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Aim:

To write a menu driven C program using switch case to perform arithmetic operations for integer from m to n.

Problem Statement:

Write a menu-driven C program that accepts two positive integers, m and n, where $m \leq n$. Display the following menu and accept the user's choice. Use a switch statement to perform the selected operation:

- 1. Print the first 10 multiples of each integer from m to n.*
- 2. Print the addition table of each integer from m to n, using addends from 1 to 10.*
- 3. Print the subtraction table of each integer from m to n, using subtrahends from 1 to 10.*
- 4. Print the results obtained by dividing each integer from m to n by the integers from 1 to 10. Display the results rounded to two decimal places.*
- 5. Exit the program.*

❖ Algorithm:

- 1) Start.
- 2) Declare m, n, choice as integers.

3) Read the input m and n.

4) Check if $m \leq n$, if not then display a message stating can't run the program and terminate the program.

5) Display the menu containing following choice:

1. Print the first 10 multiples of each integer from m to n.
2. Print the addition table of each integer from m to n, using addends from 1 to 10.
3. Print the subtraction table of each integer from m to n, using subtrahends from 1 to 10.
4. Print the results obtained by dividing each integer from m to n by the integers from 1 to 10. Display the results rounded to two decimal places.
5. Exit the program.

6) Read the user choice.

7) Use a switch statement to perform the selected operations.

7.1) choice 1: For each integer i from m to n, use another loop from j = 1 to 10 and display $i \times j$.

7.2) choice 2: For each integer i from m to n, use another loop from j = 1 to 10 and display $i + j$.

7.3) choice 3: For each integer i from m to n, use another loop from j = 1 to 10 and display $i - j$.

7.4) choice 4: For each integer i from m to n, use another loop from j = 1 to 10, calculate i / j using floating-point division, and display the result rounded to two decimal places.

7.5) choice 5: Display an exit message and terminate the menu loop.

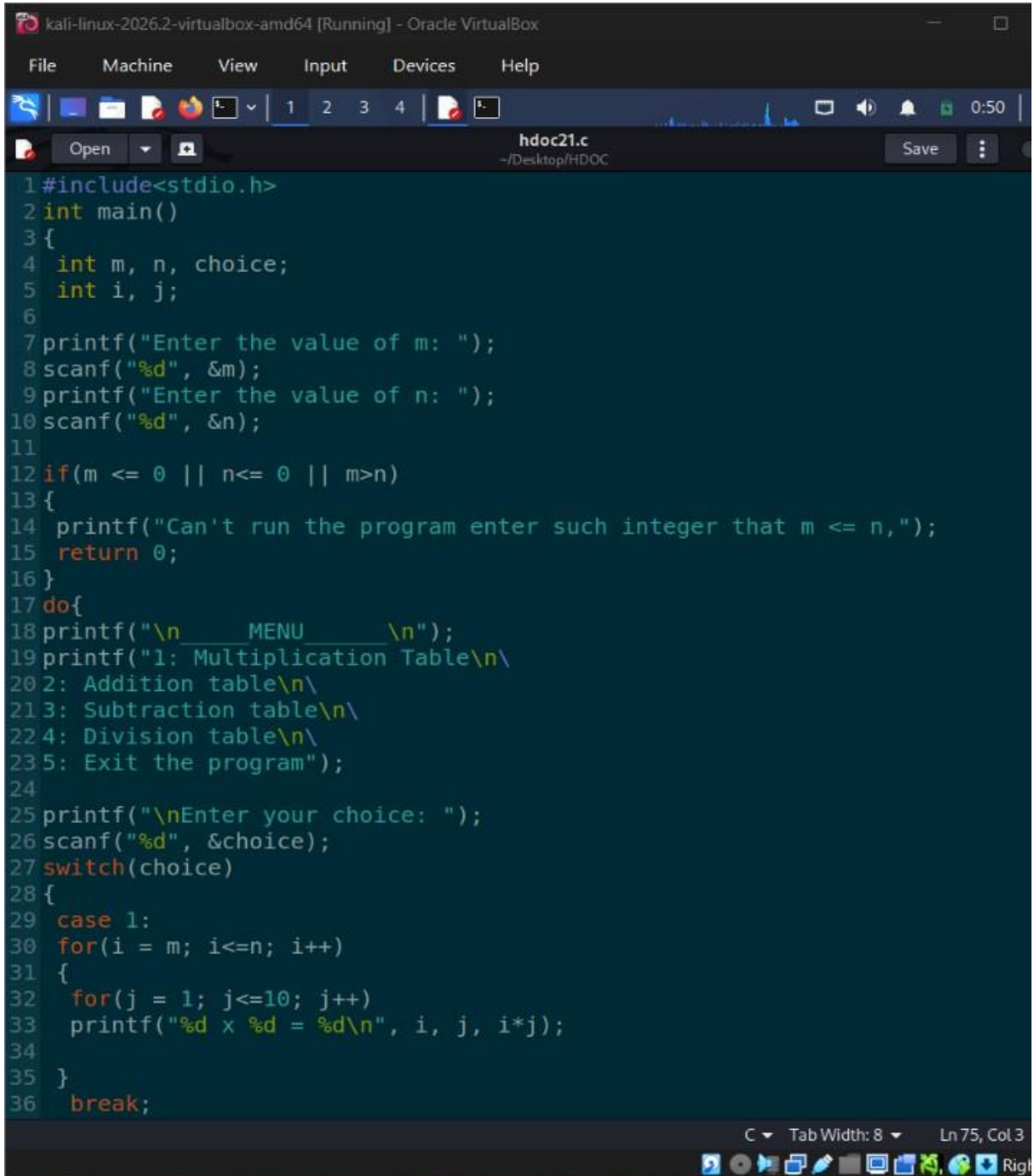
7.6) Default: Display "Invalid menu choice".

8) Stop.

Test Case Table:

Test Case	Choice	Input	Expected output	Actual output	Result
1.	1	M = 2 N = 5	Displays first 10 multiples of 2,3,4 and 5.	Displays first 10 multiples of 2,3,4 and 5.	Pass
2.	2	M = 3 N = 4	Displays addition table of 3 and 4 using addends 1-10	Displays addition table of 3 and 4 using addends 1-10	Pass
3.	3	M = 1 N = 6	Displays subtraction table of 1,2,3,4,5 and 6 using subtrahends 1-10	Displays subtraction table of 1,2,3,4,5 and 6 using subtrahends 1-10	Pass
4.	4	M = 5 N = 6	Displays division result of 5 and 6 by 1-10.	Displays division result of 5 and 6 by 1-10.	Pass
5.	5	M = 3 N = 7	Displays program is finished.	Displays program is finished.	Pass

C Program:

The image shows a screenshot of a C program being edited in a virtual machine environment. The window title is 'kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox'. The menu bar includes 'File', 'Machine', 'View', 'Input', 'Devices', and 'Help'. The toolbar shows icons for file operations and a tab labeled '1'. The file name is 'hdoc21.c' located at '~/Desktop/HDOC'. The code is as follows:

```
1#include<stdio.h>
2int main()
3{
4    int m, n, choice;
5    int i, j;
6
7    printf("Enter the value of m: ");
8    scanf("%d", &m);
9    printf("Enter the value of n: ");
10   scanf("%d", &n);
11
12   if(m <= 0 || n<= 0 || m>n)
13   {
14       printf("Can't run the program enter such integer that m <= n,");
15       return 0;
16   }
17   do{
18       printf("\n____MENU____\n");
19       printf("1: Multiplication Table\n\
202: Addition table\n\
213: Subtraction table\n\
224: Division table\n\
235: Exit the program");
24
25       printf("\nEnter your choice: ");
26       scanf("%d", &choice);
27       switch(choice)
28       {
29           case 1:
30               for(i = m; i<=n; i++)
31               {
32                   for(j = 1; j<=10; j++)
33                       printf("%d x %d = %d\n", i, j, i*j);
34               }
35           }
36       break;
```

The status bar at the bottom indicates 'C', 'Tab Width: 8', and 'Ln 75, Col 3'. The system tray on the right shows various icons including network, volume, and power.

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

hdoc21.c
~/Desktop/HDOC

```
34 }
35 }
36 break;
37 case 2:
38 for(i = m; i <= n; i++)
39 {
40 for(j = 1; j<= 10; j++)
41 {
42 printf("%d + %d = %d\n", i, j, i+j);
43 }
44 }
45 }
46 break;
47 case 3:
48 for(i = m; i <= n; i++)
49 {
50 for(j = 1; j<= 10; j++)
51 {
52 printf("%d - %d = %d\n", i, j, i-j);
53 }
54 }
55 }
56 break;
57 case 4:
58 float i, j;
59 for(i = m; i <= n; i++)
60 {
61 for(j = 1; j<= 10; j++)
62 {
63 printf("%d / %d = %.2f\n", i, j, i/j);
64 }
65 }
66 }
67 break;
68 case 5:
69 printf("Program is finished");
70 break;
71 }
72 default:
73 printf("Inavlid choice");
74 }
75 }
76 while(choice != 5);
77 return 0;
78 }
79 }
```

Compile the code using:

`gcc hdoc21.c -o hdoc21`

```
Session Actions Edit View Help
(kali@kali) - [~/Desktop/HDOC]
$ gedit hdoc21.c

** (gedit:45204): WARNING **: 00:49:57.377: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:45204): WARNING **: 00:49:57.377: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found

(kali@kali) - [~/Desktop/HDOC]
$ gcc hdoc21.c -o hdoc21
```

Execute the code using:

`./hdoc21`

Here are some screenshots of execution of code:

```
(kali@kali) - [~/Desktop/HDOC]
$ ./hdoc21
Enter the value of m: 3
Enter the value of n: 7

____MENU____
1: Multiplication Table
2: Addition table
3: Subtraction table
4: Division table
5: Exit the program
Enter your choice: 5
Program is finished

(kali@kali) - [~/Desktop/HDOC]
```

 26°C
Mostly cloudy

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VM
File Machine View Input Device

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc21
Enter the value of m: 2
Enter the value of n: 5

MENU
1: Multiplication Table
2: Addition table
3: Subtraction table
4: Division table
5: Exit the program
Enter your choice: 1
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30
4 x 1 = 4
4 x 2 = 8
4 x 3 = 12
4 x 4 = 16
4 x 5 = 20
4 x 6 = 24
4 x 7 = 28
4 x 8 = 32
4 x 9 = 36
4 x 10 = 40
5 x 1 = 5
5 x 2 = 10
```

```
kali-linux-2026.2-virtualbox-amd64
File Machine View

Session Actions Edit View Help
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30
4 x 1 = 4
4 x 2 = 8
4 x 3 = 12
4 x 4 = 16
4 x 5 = 20
4 x 6 = 24
4 x 7 = 28
4 x 8 = 32
4 x 9 = 36
4 x 10 = 40
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

MENU
1: Multiplication Table
2: Addition table
3: Subtraction table
4: Division table
5: Exit the program
Enter your choice: 5
Program is finished
```

```
kali-linux-2026.2-virtualbox-amd64 [Running]
File Machine View Input
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc21
Enter the value of m: 3
Enter the value of n: 4

MENU
1: Multiplication Table
2: Addition table
3: Subtraction table
4: Division table
5: Exit the program
Enter your choice: 2
3 + 1 = 4
3 + 2 = 5
3 + 3 = 6
3 + 4 = 7
3 + 5 = 8
3 + 6 = 9
3 + 7 = 10
3 + 8 = 11
3 + 9 = 12
3 + 10 = 13
4 + 1 = 5
4 + 2 = 6
4 + 3 = 7
4 + 4 = 8
4 + 5 = 9
4 + 6 = 10
4 + 7 = 11
4 + 8 = 12
4 + 9 = 13
4 + 10 = 14

MENU
1: Multiplication Table
2: Addition table
3: Subtraction table
4: Division table
5: Exit the program
Enter your choice: 5
Program is finished
```

```
kali-linux-2026.2-virtualbox-amd64 [Running]
File Machine View Input
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc21
Enter the value of m: 1
Enter the value of n: 6

MENU
1: Multiplication Table
2: Addition table
3: Subtraction table
4: Division table
5: Exit the program
Enter your choice: 3
1 - 1 = 0
1 - 2 = -1
1 - 3 = -2
1 - 4 = -3
1 - 5 = -4
1 - 6 = -5
1 - 7 = -6
1 - 8 = -7
1 - 9 = -8
1 - 10 = -9
2 - 1 = 1
2 - 2 = 0
2 - 3 = -1
2 - 4 = -2
2 - 5 = -3
2 - 6 = -4
2 - 7 = -5
2 - 8 = -6
2 - 9 = -7
2 - 10 = -8
3 - 1 = 2
3 - 2 = 1
3 - 3 = 0
3 - 4 = -1
3 - 5 = -2
3 - 6 = -3
3 - 7 = -4
3 - 8 = -5
3 - 9 = -6
3 - 10 = -7
4 - 1 = 3
4 - 2 = 2
4 - 3 = 1
```

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VM
File Machine View Input Devices

Session Actions Edit View Help
3 - 8 = -5
3 - 9 = -6
3 - 10 = -7
4 - 1 = 3
4 - 2 = 2
4 - 3 = 1
4 - 4 = 0
4 - 5 = -1
4 - 6 = -2
4 - 7 = -3
4 - 8 = -4
4 - 9 = -5
4 - 10 = -6
5 - 1 = 4
5 - 2 = 3
5 - 3 = 2
5 - 4 = 1
5 - 5 = 0
5 - 6 = -1
5 - 7 = -2
5 - 8 = -3
5 - 9 = -4
5 - 10 = -5
6 - 1 = 5
6 - 2 = 4
6 - 3 = 3
6 - 4 = 2
6 - 5 = 1
6 - 6 = 0
6 - 7 = -1
6 - 8 = -2
6 - 9 = -3
6 - 10 = -4

MENU
1: Multiplication Table
2: Addition table
3: Subtraction table
4: Division table
5: Exit the program
Enter your choice: 5
Program is finished

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc21
```

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VM
File Machine View Input Devices

Session Actions Edit View Help
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc21
Enter the value of m: 5
Enter the value of n: 6

MENU
1: Multiplication Table
2: Addition table
3: Subtraction table
4: Division table
5: Exit the program
Enter your choice: 4
4 / 2135048492 = 5.00
1024 / 2135048492 = 5.00
1024 / 2135048492 = 5.00
1024 / 2135048492 = 5.00
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1024 / 2135048492 = 5.00
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MENU
1: Multiplication Table
2: Addition table
3: Subtraction table
4: Division table
5: Exit the program
Enter your choice: 5
Program is finished

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc21
```